PULLBACK ALONG A MORPHISM WITH SURJECTIVE FLAT SOURCE IS FAITHFUL ON DESCENT DATA

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry 3 navigation stubs; the factorization that makes the two citations applicable is written out in the span.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Editorial note. The canonical tag of this label does not occur among the tags of the published items of descent.tex, so it is not a published duplicate.

Allowed standard prerequisites (editorial).

- Pullback along a morphism with a section is an equivalence of categories of descent data. Same chapter, Lemma lemma-morphism-with-section-equivalence, invoked at proof lines 8925-8927. Supplies the statement for the first factor.
- Pullback along a surjective flat morphism is faithful on descent data. Same chapter, Lemma lemma-faithful, invoked at proof lines 8928-8929. Supplies the statement for the second factor.
- The pullback functor of descent data. Same chapter, Lemma lemma-pullback, named in the statement at line 8918. Supplies the functor whose faithfulness is proved.

Source reference. Original lemma 0240: sources/stacks-descent/pinned-original-source.tex, source lines 8758–8758. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0243: sources/stacks-descent/pinned-original-source.tex, source lines 8897–8897. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 023Y: sources/stacks-descent/pinned-original-source.tex, source lines 8496–8496. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : X \to X'$ be a morphism of schemes over a base scheme S. Assume $X \to S$ is surjective and flat. Then the pullback functor of Lemma 023Y is a faithful functor from the category of descent data relative to X'/S to the category of descent data relative to X/S.*

Proof. We may factor $X \to X'$ as $X \to X \times_S X' \to X'$. The first morphism has a section, hence induces an equivalence of categories of descent data by Lemma 0243. The second morphism is surjective and flat, hence induces a faithful functor by Lemma 0240. □